RIN RAY

TEACHING STATEMENT

TEACHING	**5.65/6** students' rating for interest in teaching, honors linear algebra	**6 years** leading discussion sections at Northwestern	**REU lectures** weekly, University of Chicago	**12 seminars** organized, coaching every speaker	**~100 kids** hands-on stations, Math Meets Art, Münster
MENTORING	**Bridge program** mentor to students arriving sideways	**One-on-one** Directed Reading Program mentor	**Math Circles** South Side of Chicago, ages 6–12	**Building Cities** with young girls, for working mothers in math	**Queer in Math** events, and a panel in Leipzig
COMMUNITY	**Buddy Program** founded, for graduate students' mental health	**Students in crisis** refugee program in Münster; Ukrainian students at Northwestern	**Gelfand Prize** for contributions to the graduate community	**Topos House** helped found, still running	***Good Fibrations*** blog, and the interactive story *re-search*

As someone who entered academia as an outsider with a highly nontraditional background, I go out of my way to present mathematics in ways that are accessible to everyone. Mathematics is a visual, interactive, intuitive, and artistic experience, and most importantly it is a narrative one.

DISCOVERING IT TOGETHER

I often introduce the material as though we, the class, are discovering it together. We begin with examples and historical motivation, and only from there move to definitions and theorems. This recapitulates how mathematics was discovered historically. I want the students to have this same experience. Reading the work of Euler, we see that he is playing with examples until he states at the very end the theorem he deduced from those emergent patterns. Through this direct interaction with the object of interest, students can see the patterns that lead to understanding.

Concretely, I have started classes with a pile of polyhedra: cubes, tetrahedra, a soccer ball. Students count vertices, edges and faces, and we fill in a table on the board until someone notices that $V - E + F$ is always 2. Then I hand them a polyhedron with a hole through it, shaped like a picture frame, and the pattern breaks: now it is 0. Only after the class has found the rule, and found where it fails, do we write down the theorem and ask what a proof would have to explain. The definitions then arrive as answers to questions the students have already asked. Other times we have added $1 + 2 + \cdots + n$, then the squares, then the cubes, and guessed formulas from the tables, until someone noticed that the sum of the first n cubes is the square of the sum of the first n numbers. Pushing further, strange numbers appeared in the formulas: the Bernoulli numbers, the same numbers that lead, in my own research, to the prime 691.

ONE IDEA, A WHOLE CHAPTER

I like to have the students learn by doing, and then help them unite their observations into a coherent whole. As an example, when I teach linear algebra I like to begin by connecting it to image processing. I want the students to understand how a matrix affects the size, position, and rotation of images. To do so, I have my students use the explorable explanation *The Magnificent 2D Matrix* by my friend and master expositor Nicky Case, which is publicly available alongside many other similarly accessible tales through their website ncase.me. When the students change the 2D matrix, they can see what the matrix does visually. This requires no mathematical background. Once they understand intuitively the transformative properties of a matrix, I like to introduce the concept of a determinant of a matrix M as the volume of a unit cube transformed by M. I then introduce the defining properties that characterize the absolute value of the determinant visually; for example, swapping two rows leaves the size of the volume unchanged, just as swapping "width" and "height" does, and only flips its orientation, which is why the determinant changes sign. In this way, they can survey an entire system of discoveries that follow from a single essential idea, volume, which, in their logical order, harmony, and brilliance, appear transparent and as an organic whole, like a painting before the mind's eye.



a sketch of mine on exactness

WITH HISTORY, AND WITH PICTURES

As an example of presenting material with historical context, when I am teaching real analysis, I split my exercises between the standard book, Rudin, and the corresponding chapter in *Analysis by Its History*. I also make sure to provide visual textbooks when possible, such as *Visual Complex Analysis*.

THE PERSON IN FRONT OF ME

I cannot know every student in a large course, but I can find out who is in the room. Early on I ask what everyone is studying and what they love outside of class, and I bring it back into the material. Students arrive with their own reasons: a musician who wants to know knot theory, a chemist who wants to understand the wallpaper groups behind crystals. The theorem is the same for everyone, but once it answers their question, it belongs to them. I also bring props, like pipe cleaners to tie into knots and untangle, and 3D-printed surfaces to explain Morse theory, so that students can hold an idea in their hands before they have to write it down.

When a section is full of future biologists, psychologists and doctors, I teach p-hacking: what happens when you keep slicing the data until something comes out significant. These students will run experiments and read clinical studies for the rest of their careers, in fields where a large share of published findings do not replicate, partly for exactly this reason. Knowing why twenty tries at $p < 0.05$ will usually turn up one false discovery is the difference between doing science and fooling yourself. Teaching it early, to the students who will run tomorrow's experiments, is one of the most direct ways I know to raise a generation of scientists whose results hold up.

I also really enjoy working with students one-on-one, responding to their individual needs. For example, I have provided my Math Circles, REU, and DRP students with: sets of puzzles to play with, objects of direct interest, a question to guide their play, or sequential learning of concepts guided by resources on my website rin.io/resources. This works for students of all levels, from beginners to graduate students.

TEACHING EXPERIENCE

In the Classroom. I have done every part of teaching a college course except being the name on the syllabus. For six years at Northwestern I led discussion sections for Intensive Linear Algebra and Multivariable Calculus (the honors sequence), Linear Algebra, and Discrete Mathematics, and wrote quizzes and their solutions; at the University of Chicago REU I gave weekly lectures. I am ready to run a course of my own. In my 2021 evaluations for the honors sequence, students rated me 5.65 out of 6 for interest in teaching and 5.48 for clarity. A few of their comments:

> *"She effectively helps students in understanding the material better and clearly explores subtle nuances to the material covered."*
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> *"Rin was very helpful and really great to have as a TA. They made it such an inclusive, comfortable space for me."*
>
> *"I looked forward to going to discussion sections because they would always be happy to see us and happy to help."*
>
> *"She cares a lot about students and gives feedback in an encouraging way. She is super helpful and shares interesting things about what happens after math classes."*
>
> *"Very efficient in getting grades back to the students. Also very understanding of the workload of students outside of the class and of students in general, doesn't really have that 'god TA complex'."*
>
> a sample of student comments, honors linear algebra and multivariable calculus, Spring 2021

Mathe trifft Kunst. For the 2025 Day of Mathematics, run by the Mathematics Münster Cluster of Excellence, I designed hands-on stations, and the picture prompts for them, for about a hundred children. The children found the symmetries in Escher's drawings and worked out how he made them, then cut and taped their own tiles from triangles and traced them into tessellations, including a giant tessellation outside. They figured out dual polytopes for themselves, drew Sierpiński triangles from a printed first step and fitted them together into Sierpiński hexagons, and held rule-of-thirds and golden-triangle overlays over photographs to see how the pictures were composed. I gave the instructions as a series of pictures with only a few words, because children can be trusted to work things out.

Seminars and Conferences. I also teach graduate students and colleagues through learning seminars, and in every seminar I run, I coach the speakers on giving their talks. At Northwestern I organized seminars on factorization homology, p-adic geometry (Geometry for Prime Addicts), formal moduli problems, the Tamagawa conjecture for function fields (TamagaWHAT), applications of topos theory, moduli spaces, and computation in number theory and homotopy theory (Stop Staring and Compute!). In Münster I have organized working groups on the K-theory of the integers, Euler characteristics and zeta functions, K-theoretic Poitou–Tate duality, linearity in p-adic Lie theory, and duality and linearization for p-adic Lie groups. In 2024 our homotopy working group ran as a feedback seminar: we learned together what makes feedback useful or useless, and every talk had assigned feedback-

givers. In 2024 I was on the organizing committee of the Young Topologists Meeting in Münster, a conference run by and for PhD students and postdocs. I made the meeting easy to take part in, with throwable plush microphone cubes so anyone could ask a question, a quiet room, and an art section; I also drew its logo and designed the schedule with Natalia Pacheco-Tallaj.

EXTRACURRICULARS

Mental Health Support. I encourage students to form healthy boundaries and establish their work-life balance to prevent burnout. The graduate students at Northwestern who entered during the pandemic did not form social groups. To combat this, in 2021 I founded a funded program, the Mental Health Math Buddy Program, pairing first- and second-year graduate students with older graduate students to meet once a quarter for mental health check-ups. I was also a TGS Peer Mentor in the Office of Diversity and Inclusion, which helps first-year PhD students from diverse backgrounds become acclimated to Northwestern. Toward supporting the mental health of graduate students, I established the Graduate Life Appreciation Fund for enriching the climate of the graduate program at Northwestern, and served as its treasurer. This fund helps students participate in activities they might not be able to afford otherwise.

Mentoring the Real Outsiders. Much of my teaching happens with people who come to mathematics sideways: late, from another field, or from circumstances nobody planned for, and they tend to be sent to me. Through the Causeway Bridge Program, a year-long program that brings students from underrepresented groups into the mathematical sciences, I mentored students with the wildest paths in: a rich and interesting group, usually from complex circumstances, often arriving by way of physics, computer science, or entrepreneurship. Having a background in all three myself, I help them get acquainted: with the culture, the unspoken rules, and the people, until our research environment feels like theirs too. I build the doors I wish had been there, and then I stand at them and say come in.

Students in Crisis. When the war in Ukraine began in 2022, I fought Northwestern's administration, and won, so that Ukrainian students, who had to stay registered full-time to keep their visas, could take incompletes and be given more leniency. In October 2023 I set up a program at the University of Münster for refugee PhD students, hosting them for up to six months and waiving the usual processing time so they could leave at once.

From Engineers to Laypeople. For engineers and scientists, I helped found and fund the Topos House, which hosts 6–12 scientists in a central house dedicated to creating the optimal conditions for interdisciplinary work focused around mathematics. I also connect people to Davide Radaelli and Nic Ford, two people teaching undergrad and grad-level math to engineers, nicf.net/about. I am also making *re-search*, a story about the joy of understanding a concept for the first time, told in writing with interactive pieces the reader can play with, for people with no mathematics at all.

Expository Blog. It's borderline impossible to get comfortable with many of the tools in modern mathematics without having a professor that wants to spend a decent amount of time working with you. This is inaccessible to most people. My blog *Good Fibrations* aims to bridge that gap. Here is an example of a review of my expository style by someone from a liberal arts college who used it as a way to prepare for graduate school during the pandemic:

> *"I really like your drawings and I really really like your chatty, fun happy writing style which makes learning feel more safe and less hostile."*

Getting In, Staying In. My resources page, rin.io/resources, collects what I wish someone had handed me: a mental health section that starts with the person, the visual books and interactive explanations I teach from, and a guide to the ways into mathematics and science for people who arrive late or sideways, from bridge programs to a PhD to funding and residencies open to everyone.

Supporting Parents. From 2017 to 2023 I ran Building Cities with Young Girls in Evanston: every week, girls built forts and cityscapes with me, giving working mothers in math some time back.

Queer in Math. I am visibly queer and alt, and being fully myself at work makes it easier for students, and even seasoned mathematicians, to be themselves too. I go to Queer in Math events, and in 2024 I spoke on a panel at Queer in Math Day in Leipzig.

> *I strive to make sure people with any background can get acquainted with both the methodology of mathematics and the immersive feeling of doing mathematics.*